Skip List Music Manager

1. PROJECT OVERVIEW

Team Members

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TA or Instructor Consultation

Wilhelmina, spoke during lab on 10/21

GitHub Repository

https://github.com/wakeupdotexe/CSC212-Skip-List.git

2. PROBLEM AND MOTIVATION

Our project will address the issue of quickly searching through a linked list. This problem is interesting because with issues of sorting, binary searches are usually a reliable way of finding an item efficiently. However, binary searches within singly-linked lists seem to be only as efficient as a linear search. We would like to find a more efficient solution.

3. PROPOSED SOLUTION AND DATA STRUCTURE

The application we will create will be a music player and manager that utilizes a skip list for search functioning. A skip list will allow for more efficient searching through a singly-linked list of files, saving a lot of time when looking for an item at the end of a very long list.

4. BASELINE AND EVALUATION PLAN

Our baseline implementation will be a binary search. I plan to use a sample of my personal music collection as input data (consisting of about 130 files). I will randomly select song titles to use as test cases, as well as some titles not included in the input data. We will take note of running time and amount of operations for the sake of comparison.

5. PLANNED DEMONSTRATION

For our live demo, we plan to have an application that accesses a user's local audio files and allows the user to play and search through their music. We will display how the search function works, talk about our process implementing the skip list, and explain the differences we found between the skip list and a binary search.